

Employee Workforce Analytics

1. Project Overview and Objective

This project focuses on analysing employee and salary change data using **Microsoft Excel** and **Power BI**. The dataset is cleaned and transformed using Excel and Power Query, followed by data modelling and visualization in Power BI. The interactive dashboard provides insights into employee performance, salary trends, workforce status, and departmental distribution to support HR decision-making.

The main objective of this project is to clean, transform, and analyse HR data to identify meaningful workforce insights. Using Excel for data preparation and Power BI for interactive dashboards, the project helps visualize employee performance, salary history, hiring trends, and workforce status, enabling data-driven HR decisions.

2. Data Sources

- **Source Description and Timeline:** [The Ultimate HR & People Analytics Dataset Generator | Excelx.com](#) and 2018-2026.
 - **Domain:** Employee Performance & Salary Analysis
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3. Problem Statement

- To clean and transform raw employee and salary change data by correcting data types, handling missing values, and standardizing the dataset to ensure accurate and reliable analysis.
 - To analyse employee information such as department, job title, performance rating, employment status, and salary details to understand workforce distribution and employee performance.
 - To examine salary change history by analysing salary increments, hike percentages, and the reasons for salary revisions to identify compensation trends within the organization.
 - To perform exploratory data analysis using Excel formulas, descriptive statistics, and Pivot Tables to summarize key HR metrics and uncover meaningful business insights.
 - To develop an interactive HR Analytics Dashboard in Power BI that visualizes employee performance, salary trends, workforce status, and departmental insights to support data-driven HR decision-making.
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4. Attribute (Column /Features) Details:

EMPLOYEE :

Attribute Name	Data Type	Description
Employee ID	Text	Unique identifier for each employee
Full Name	Text	Name of the employee
JobTitle	Text	Employee's job designation
Department	Text	Department where employee works
Email	Text	Employee's email address
Gender	Text	Gender of the employee
Hire Date	Date	Date the employee joined the organization
Salary	Whole Number	Current salary of the employee
Manager ID	Text	ID of the employee's reporting manager
Performance Rating	Decimal	Employee performance rating (out of 5)
Status	Text	Employment status (Active/Terminated)

SALARY CHANGES:

Change ID	Text	Unique identifier for each salary change
Employee ID	Text	Employee identifier linked to the Employee Data table
Change Date	Date	Date of the salary revision
Old Salary	Whole Number	Previous salary of the employee
New Salary	Whole Number	Updated salary after revision
Hike %	Percentage	Percentage increase in salary
Reason	Text	Reason for the salary change (Merit/Promotion)

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
 - **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.
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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
 - **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.
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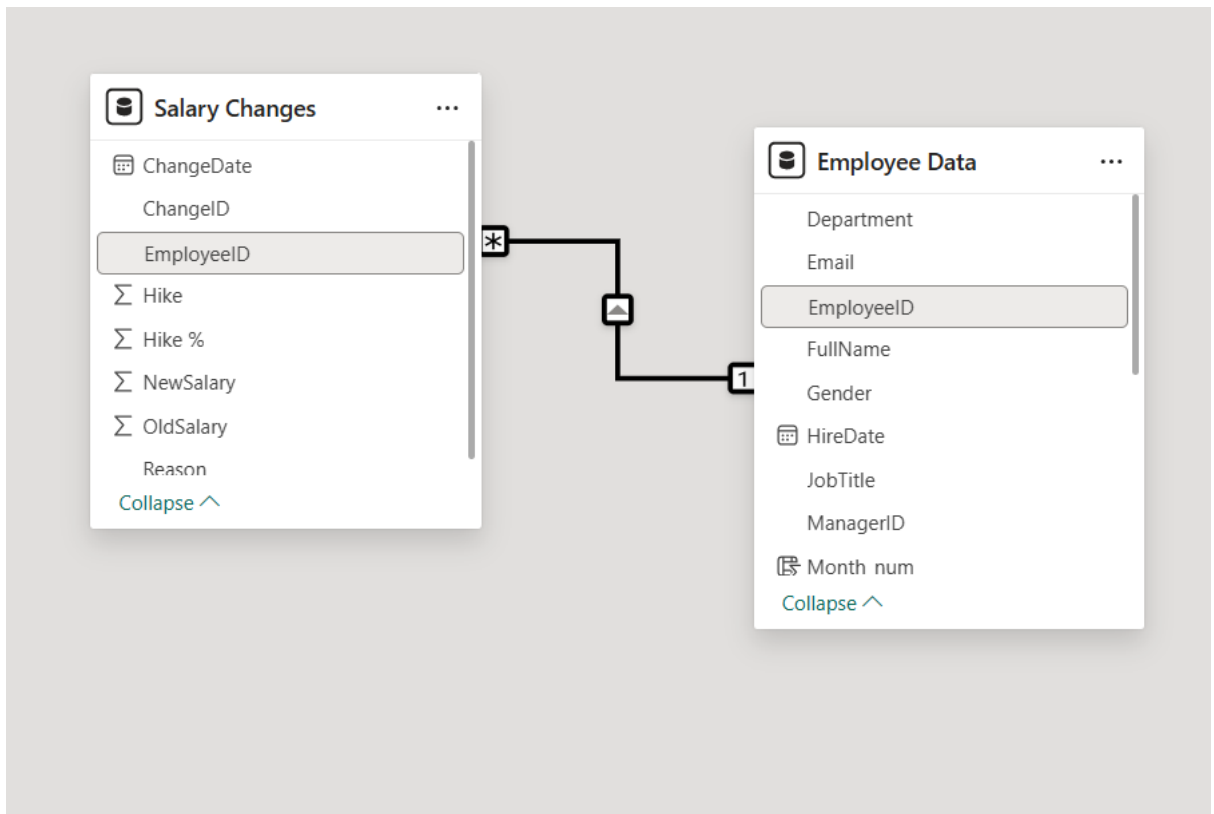
7. Data Modelling and DAX (Power BI)

- **Data Model:**

The data model was created by establishing a relationship between the **Employee Data** and **Salary Changes** tables using the **Employee ID** field. This relationship enables accurate analysis of employee information and salary changes across the dashboard.

Relationship Details

- **Primary Table:** Employee Data
- **Related Table:** Salary Changes
- **Relationship Key:** Employee ID
- **Cardinality:** One-to-Many
- **Cross Filter Direction:** Single



DAX Measures Implemented

The following DAX measures were created to support the dashboard analysis:

- Total Employees

Total Employees = `COUNT('Employee Data'[EmployeeID])`

- Active Employees = `CALCULATE(`
`Count('Employee Data'[EmployeeID]),`
`'Employee Data'[Status]="ACTIVE")`

- Terminated Employees = `CALCULATE(`
`Count('Employee Data'[EmployeeID]),`
`'Employee Data'[Status]="TERMINATED")`

- Average Hike (%) = `AVERAGE('Employee Dataset'[Salary])`

8. Analysis and Visualizations (Power BI)

Visualizations Used in Dashboard

- KPI Cards
 - Total Employees

- Active Employees
- Terminated Employees
- Average Hike Percentage
- **Donut Chart**
 - Employee Distribution by Gender
- **Pie Chart**
 - Employee Status (Active vs. Terminated)
- **Clustered Bar Chart**
 - Average Salary by Department
- **Clustered Bar Chart**
 - Employee Count by Department
- **Line Chart**
 - Hiring Trend by Year
- **Clustered Bar Chart**
 - Average Performance Rating by Department
- **Tables**
 - Department-wise Performance Summary
 - Year-wise Hiring Summary

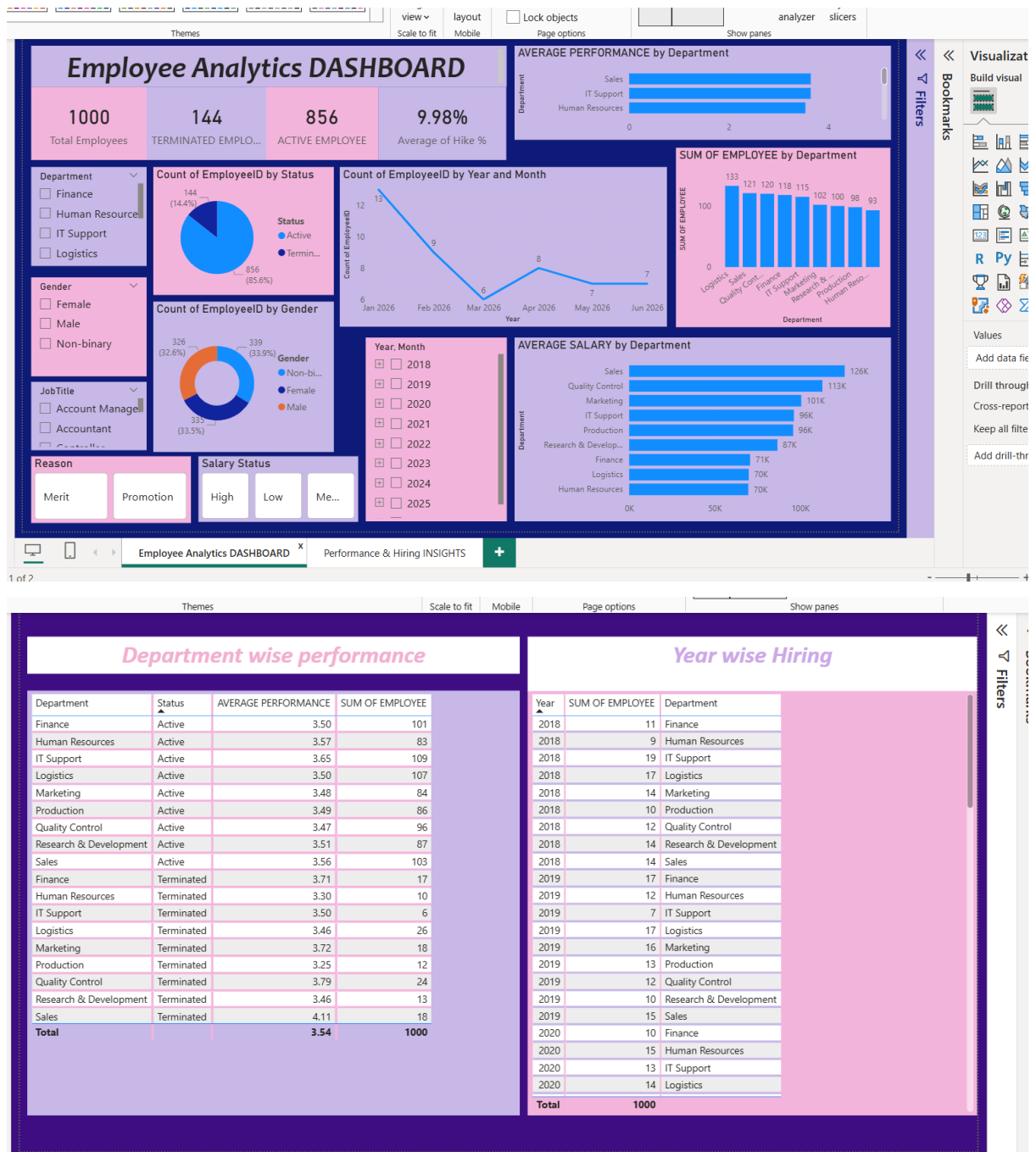
Interactive Features

- Department Slicer
- Gender Slicer
- Job Title Slicer
- Salary Status Slicer
- Reason Slicer
- Year Slicer

Dashboard Layout

The dashboard is divided into two pages:

- **Employee Analytics DASHBOARD:** Workforce Overview & Performance Analysis
- **Department & Hiring INSIGHTS:** Department Wise Performance & Year Wise Hiring



9. Insights & Conclusions

Key Findings:

- The organization has **1,000 employees**, of which **856 (85.6%) are active** and **144 (14.4%) are terminated**, indicating a stable workforce.
- The **average salary hike is 9.98%**, reflecting the company's compensation trends.
- The workforce has a **balanced gender distribution**, promoting diversity across the organization.
- Sales, Logistics, and Quality Control** have the highest employee count, indicating these are the major operational departments.

- The **Sales department** has the highest average salary, while **Finance and Human Resources** have comparatively lower average salaries.
 - The **average performance rating across all departments is below 4 out of 5**, indicating that employees perform consistently but there is still scope for improvement to achieve higher performance levels.
 - Salary increments are mainly driven by **Merit** and **Promotion**, showing that the organization follows a performance-based reward system.
 - The year-wise hiring analysis highlights recruitment patterns and helps HR understand workforce growth over different years.
- **Provide the analysis insights:**
- Total workforce consists of **1,000 employees**, with **856 active** and **144 terminated** employees.
 - The **average salary hike is 9.98%**.
 - **Sales, Logistics, and Quality Control** have the highest employee count.
 - The **Sales department** has the highest average salary.
 - The **average performance rating in all departments is below 4 out of 5**, indicating room for improvement.
 - Year-wise hiring analysis provides a clear view of recruitment trends and workforce growth over different years.
 - Salary differences are influenced by job roles and department responsibilities.
 - Most salary hikes are based on **Merit** and **Promotion**.
 - Performance below 4 suggests the need for additional training and performance improvement initiatives.
 - If the current hiring and retention trends continue, the organization is expected to maintain a stable workforce.
 - Merit-based salary hikes and promotions are likely to continue motivating employees and supporting career growth.
 - Without focused employee development programs, the average departmental performance is likely to remain below **4 out of 5**.

- Continuous monitoring of hiring, salary, and performance trends will help HR anticipate future workforce requirements
 - Continue implementing **performance-based salary hikes and promotions** to reward deserving employees.
 - Provide **training, mentoring, and skill development programs** to improve employee performance across all departments.
 - Introduce **performance-based recognition programs** to motivate and encourage employees.
 - Set **clear Key Performance Indicators (KPIs)** for each department to monitor and improve performance.
 - Offer **coaching and regular feedback** to employees with lower performance ratings.
 - Use the **HR Analytics Dashboard** regularly to support data-driven decision-making, monitor workforce performance, and improve HR planning
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10. Conclusions

The integration of **Excel and Power BI** successfully transformed raw HR data into meaningful insights through data cleaning, analysis, and interactive dashboards. The project helps monitor employee performance, salary trends, hiring patterns, and workforce distribution, enabling informed HR decision-making and better organizational planning.